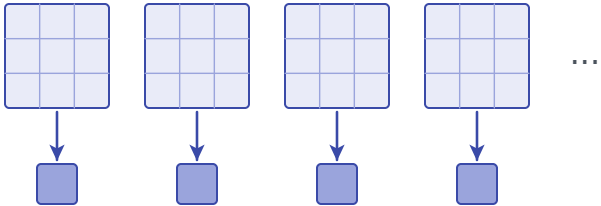
Stage 1: C = 32, so the middle is 128 channels wide. Tokens in = tokens out; the 3 x 3 window is what tells tokens where they are.

Depthwise: each channel has its own 3 x 3 filter



channel 1, 2, 3, 4 ... each mixes only its own 8 neighbours

Parameters at 128 channels

ordinary 3x3: 128 x 128 x 9 + 128 = 147,584

depthwise 3x3: 128 x 9 + 128 = 1,280

115 times cheaper; channel mixing is left to the two Linear layers on either side.